INCH-POUND

MIL-A-82769(OS)
29 December 1989
SUPERSEDING
WS 22381B
9 December 1986
(See 6.5)

MILITARY SPECIFICATION

ASSEMBLY, AFT CLOSURE, SILICONE RUBBER, ROOM TEMPERATURE VULCANIZING, IN-PROCESS REQUIREMENTS FOR

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the in-process requirements for one type of room temperature vulcanizing (RTV) silicone rubber aft closure assembly, referred to herein as the assembly (see 6.1). The in-process requirements herein cover assembly characteristics which must be verified prior to assembly and fabrication of the end product.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1340

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SPECIFICATIONS

FEDERAL

TT-I-735 Isopropyl Alcohol

STANDARDS

MILITARY

MIL-STD-453 Inspection, Radiographic

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 740 Methyl Ethyl Ketone (DoD) adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

UNITED STATES PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia Absorbent Gauze

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 In-process. The assembly shall conform to the requirements specified herein.

3.2 Surface condition. Defects in surface condition of the assembly shall not exceed the limits specified in table I.

TABLE I. Surface defect conditions.

Component	Defect	Permissible Condition Extent
Throat inlet, closure insulation, throat support and approach insert	Blemishes (see 6.3.1)	Shall not exceed one square inch, 2.0 inches in length or 0.03 inch in depth. No two greater than 0.5 square inch or 0.03 inch in depth shall be closer together than two inches. No more than three blemishes per part that violate the criteria.
	Wrinkled areas (see 6.3.4)	None allowed.
	Delaminations (see 6.3.3)	None allowed.
	Cracks (see 6.3.2)	None allowed.
Aft closure shell O" ring lockwire groove, threads, major O D., and throat insert	Blemishes	None allowed.
Other coated surfaces of aft closure shell	Blemishes	None deeper than 0.002 inch.
Interface bond	Voids (see 6.3.10)	None larger than 0.200 inch circumferentially by 0.150 deep by full bond width. Accumulative voids less than 30 percent of circumferential bond length.

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3.3 **Subsurface condition.** Defects in subsurface condition of the assembly shall not exceed the limits specified in table II.

TABLE II. Subsurface defect limits.

Defect	Permissible Extent
a. Cracks (see 6.3.2)	None allowed.
b. Low density line (see 6.3.5)	None greater than 0.050 inch wide. No more than 5 per part. (A low density line between the same two fabric plies which extends over part circumference is considered a single line.
c. Low density areas (see 6.3.6)	Any single area less than 0.100 inch in length or 0.100 inch in width. No more than one in any 1 square inch area. No more than 5 per part.
d. High density inclusions (see 6.3.7)	Any single inclusion shall be less than 0.050 inch in length or width in the plane of the radiograph. Total surface area of inclusions shall be less than 0.025 square inch in any square inch.
e. Delaminations (see 6.3.3)	None allowed.

3.4 **Interface bonds.** Separations, unbonds and voids in the interface bonds shall not exceed the limits specified in table III (see figure 1.)

3.5 **Workmanship.** The assembly shall be free from surface contaminants, grease, oil, foreign material, or other defects which could render the part unsuitable for the intended use.

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TABLE III. Interface bond defect criteria.

Component 1/	Permissible Extent (As measured on a single X-ray film or video tape tangential radiograph)
1. Closure shell to:	
a. Throat support	No single separation or void length greater than 0.25 inch. Accumulative separations and voids less than 20 percent of the bond length.
b. Throat inlet	No single separation or void length greater than 0.30 inch. Accumulative separations and voids less than 20 percent of the bond length. Accumulative porosity, separations and voids, less than 35 percent of the bond length.
c. Approach insert	No single separation or void length greater than 0.20 inch. Accumulative separations and voids less than 20 percent of the bond length. Accumulative porosity, separations and voids, less than 35 percent of the bond length.
d. Closure insulation	No single separation or void length greater than 0.70 inch. Accumulative separations and voids less than 20 percent of the bond length. Accumulative porosity, separations and voids, less than 35 percent of the bond length.
2. Approach insert to: a.) Throat inlet b.) Closure insulation	No single separation or void length greater than 0.15 inch. Accumulative separations, voids and porosity less than 25 percent of the bond length.

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TABLE III. Interface bond defect criteria (Continued).

Component 1/	Permissible Extent (As measured on a single X-ray film or video tape tangential radiograph)
3. Throat inlet to: a.) Throat support	No single separation or void length greater than 0.10 inch. Accumulative separations, voids and porosity less than 20 percent of the bond length.
b.) Throat insert	No single separation or void length greater than 0.30 inch. Accumulative separations, voids and porosity less than 20 percent of the bond length.
4. Throat insert to: a.) Throat support	No single separation or void greater than 0.25 inch. Accumulative separations and voids less than 20 percent of the bond length.

1/ Refer to 6.3 for definitions of porosity (see 6.3.8), separations (see 6.3.9) and voids (see 6.3.10).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of section 3. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an

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acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actuál, nor does it commit the Government to accept defective material.

4.2 Quality conformance inspections. Quality conformance inspections shall consist of the examination and inspection listed in table IV.

TABLE IV. Quality conformance inspections.

Inspection	Requirement	Test Method
Visual examination	3.2 & 3.5	4.3.1
Radiographic inspection	3.3 & 3.4	4.3.2

4.3 Methods of inspection. The following inspection methods shall be used. Unless otherwise specified in the method, all weights, volumes, temperatures, and times shall be measured to the nearest specified unit or decimal.

4.3.1 Visual examination. Each assembly including aft closure shell, "O" ring surfaces, lockwire groove, threads, major O.D., RTV interface bonds, and throat insert shall be visually examined to determine conformance to 3.2 and 3.5. Noncompliance with any specified requirement shall constitute cause for rejection of the assembly. Methyl ethyl ketone (MEK) or isopropyl alcohol shall be used as an aid in determining conformance to surface crack and delamination requirements specified in 3.2 as follows:

a. Wet surface with MEK conforming to ASTM D 740 or isopropyl alcohol conforming to TT-I-735.

CAUTION

Minimize solvent contact with RTV bond lines.

b. Wipe off excess solvent with type VIII absorbent gauze conforming to The United States Pharmacopeia.

c. Allow part to dry in air. Defects will appear as a wetted line or area when surrounding area appears dry.

4.3.2 Radiographic inspection. Each assembly shall be radiographically inspected to determine conformance to the subsurface and interface bond requirements specified in 3.3 and 3.4. Noncompliance with any specified requirements shall constitute cause for rejection of the assembly. The inspection shall be subject to the following conditions:

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a. Film radiography shall be in accordance with MIL-STD-453 and figure 2 specified herein.

b. Real time radiography (RTR) shall be in accordance with the contractor's written procedures. The procedures shall include but not be limited to the following:

(1) System description and operation

(2) Process control, to include tests and inspections verifying video monitor resolution, grey scale range, electronic or geometrical magnification or both, and flatness of the as-viewed and recorded image.

(3) A RTR test procedure which is capable of consistently producing the desired inspection quality level. The test procedure shall include but not be limited to the following:

(a) A drawing or sketch of the component showing the location of the imaging device and penetrameter with respect to the radiation source.

(b) The rads and focal spot size of the radiation source.

(c) Radiation source to imaging device distance.

(d) Type of video recording tape.

(e) Video field of view size, camera f stop, part rotation speed and axial index increment if applicable, and positive or negative video mode.

(f) Pertinent image processing parameters such as frame averaging, kernel size and shape, and video mixing ratio.

(g) When specified (see 6.2), a means of monitoring and verifying radiation dose rate.

c. The permanent record of inspection (x-ray film or video tape) shall demonstrate a steel wire penetrameter sensitivity of 1.5 percent.

5. PACKAGING

This section is not applicable to this specification.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The assembly specified herein is intended for use as a component in the Standard Missile MK 104 Dual Thrust Rocket Motor.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1, 2.1.2 and 2.2).

c. When a means of monitoring and verifying radiation dose rate must be included in the RTR test procedure (see 4.3.2b (3)(g).

6.3 Definitions.

6.3.1 Blemishes. Any superficial imperfection or marks or deformities such as scratches, crazing, gouges, cuts, or stains.

6.3.2 Cracks. A separation which extends across fabric plies.

6.3.3 Delaminations. A separation between fabric plies which is parallel to the fabric plies.

6.3.4 Wrinkled areas. Distortion in fabric orientation normal to the plane of the fabric.

6.3.5 Low density line. A low density area between fabric plies which is parallel to the fabric plies.

6.3.6 Low density area. A low density area which has appreciable width as measured perpendicular to the plies and is not necessarily oriented in any specific manner with respect to the plies.

6.3.7 High density inclusion. Object that has higher density than that of the surrounding material.

6.3.8 Porosity. Areas of missing adhesive which do not extend the full bond width.

6.3.9 Separations. Area where adhesive has not bonded to the surrounding material.

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6.3.10 **Voids.** Areas of missing adhesive which extend the full bond width.

6.4 **Subject term (key word) listing.**

Rocket Motor, Dual Thrust Mk 104
Standard Missile

6.5 **Supersedure information.** MIL-A-82769 incorporates the following engineering change proposals (ECPs) and notices of revision (NORs):

ECPs	NORs
ARC001 (7/27/88)	ARC001.3 (7/27/88)
ARC006 (9/29/88)	ARC006.1 (9/29/88)

Preparing Activity
NAVY-OS
(Project 1340-N778)

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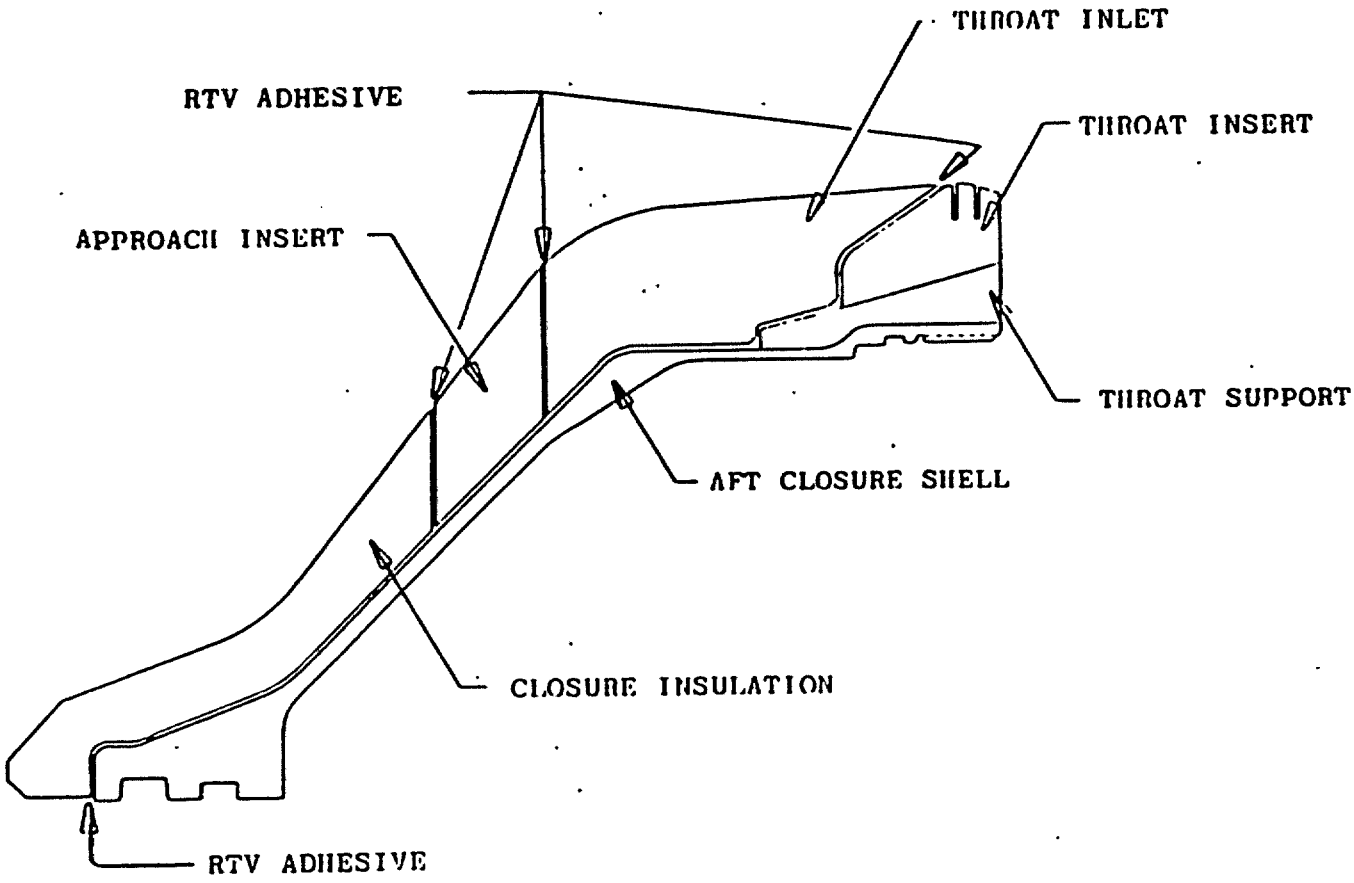
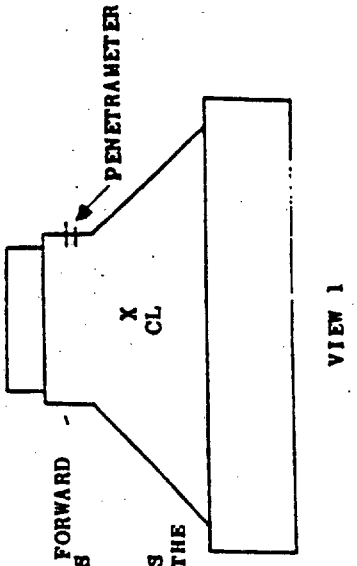


FIGURE 1. Aft closure assembly.

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NOTES

1. DEGREES SHALL BE MEASURED ON AFT FACE LOOKING FORWARD IN A CLOCKWISE DIRECTION TO PRODUCE TANGENTIALS EVERY 60°.

2. TIFY FILM, USING LEAD NUMBERS, WITH DEGREES GENT POINT AND THE LAST THREE DIGITS OF THE IAL NUMBER.

LINE SHALL BE LOCATED 3.00 +/-.10 D OF AFT END OF PART.

FIGURE 2. Radiographic inspection requirements.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-A-82769	2. DOCUMENT DATE (YYMMDD) 29 December 1989

3. DOCUMENT TITLE
ASSEMBLY, AFT CLOSURE, SILICONE RUBBER, ROOM TEMPERATURE VULCANIZING

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY NAVAL ORDNANCE STATION (CODE 3730) INDIAN HEAD, MD 20640-5000

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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